

ANDREW KWOLEK

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EDUCATION

Northwestern University

Master of Science in Robotics

- GPA: 4.0/4.0

Evanston, IL; September 2025

University of Michigan - Ann Arbor

Bachelor of Science in Engineering

- B.S.E in Mechanical Engineering
- B.S.E in Computer Science
- Cum Laude

Ann Arbor, MI; April 2022

TECHNICAL SKILLS

C++ | Python | SolidWorks | ROS2 | Go | Unity | Linux | Git | Jira | Matlab | Java | PyTorch | Java

EXPERIENCE

TikTok, Site Reliability Engineer

Seattle, WA; July 2022 - August 2024

- Maintained microservices dealing with user accounts, paid content, and creator monetization
- Managed full scale cloud service migration involving multiple teams from US and China for service leveraged by hundreds of internal users
- Initialized and deployed 2 code repositories, 7 cloud services, 2 load balancers, 2 relational databases, 3 object storage buckets, 2 API domains, 1 data exchange security channel, and 1 message queue
- Designed and developed automation tools in Golang used by teams across organizations. Reduced deployment times by ~90%
- Led code development for internal oncall integration tool used by 20+ teams throughout company

General Atomics ASI, Software Engineering Intern

Poway, CA; June 2021 - August 2021

- Developed Python test scripts for 6 autonomous flight control patterns. Merged ~800 lines of code into production repository using SVN version control software for use in full system CI/CD test suite
- Created framework in Python to bridge test scripts to human machine interface
- Designed class diagrams, logged test cases and test procedures, and developed 28 new system requirements for automated testing framework

Sarcos Robotics, Robotics Intern

Salt Lake City, UT; July 2020 - October 2020

- Optimized Force Balancing and Payload Estimation software for Guardian XO robotic exoskeleton to produce fluid movement and reduce user stress
- Wrote new technical documentation for 16 exoskeleton link test procedures used to verify all link features worked properly
- Designed link components, parts, and test fixtures in SolidWorks. Manufactured and assembled parts for use in exoskeleton performance and testing

PROJECTS

Apex Putter Golfing Robot, Motion Planning Lead

Evanston, IL; October 2024 - December 2024

- Programmed custom ROS2 package for Franka robot arm to make hole-in-one using computer vision and motion planning
- Created CAD model and 3D printed custom golf club end-effector used to putt golf ball
- Trained YOLO computer vision model to detect golf ball and hole location for accurate club placement and swing power in 10 square inch range
- Planned and executed 4 different motion paths for ready state, aligning putter next to ball, and swinging club to hit ball in correct direction
- Presented 15 minute live demonstration in front of students and faculty

Robot Chess Arm, Software Lead

Ann Arbor, MI; January 2021 - April 2021

- Designed, manufactured, and programmed robotic chess arm for mechanical engineering curriculum capstone project
- Modified and adapted existing code base in Python to control arm movement and communicate with Stockfish chess AI through an API
- Sourced actuators and hardware through motor selection practices to identify necessary torque and current
- Devised 3D model in SolidWorks consisting of machined and purchased parts. Assembled robot and demonstrated functionality by navigating arm to all 64 chess board squares

Ankle-Foot Prosthesis Design Project, Research Assistant

Ann Arbor, MI; September 2019 - December 2019

- Conducted undergraduate research on subsystem designs for robotic prosthetic ankle in Prof. Elliott Rouse's Neurobionics Lab
- Devised 3 unique designs through hand sketches, SolidWorks, and rapid prototyping. Derived equations to determine input parameters for design's motor selection and sourced parts to implement each design
- Performed Pugh analysis to determine optimal design based on 7 criteria. Delivered 20 minute presentation of findings to Robotics faculty and students at Mechanical Engineering undergraduate symposium